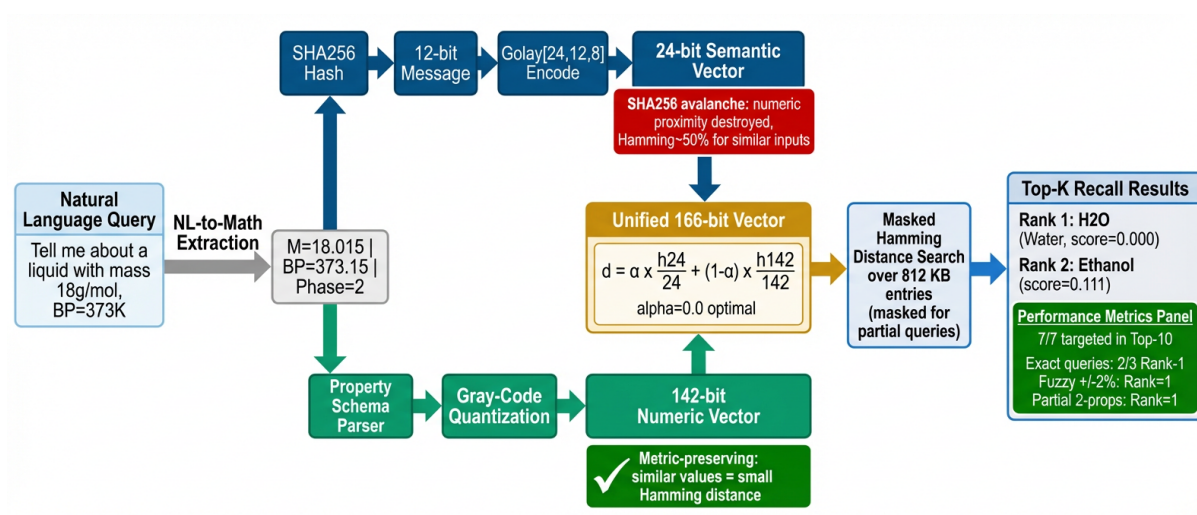


# Universal Binary Principal (UBP) Geometric Language Machine

Iterative Development and Experimental Study Report



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## Abstract

The Universal Binary Principal (UBP) system encodes knowledge as 24-bit Golay [24, 12, 8] codewords and retrieves entries via Hamming distance search. This report documents a five-step iterative experimental study aimed at enabling *deterministic*, property-based recall of knowledge-base entries from natural-language queries—building a Geometric Language Machine. Baseline diagnostics established that the incumbent SHA256-derived Golay encoding completely destroys numeric proximity (mean Hamming distance  $\approx 10.13/24$  for near-identical inputs), making property-based retrieval impossible with the original pipeline. We designed and validated a Gray-code quantised numeric vector engine (117 $\rightarrow$ 142 bits) that preserves small numeric changes as small Hamming distances. Unified 166-bit representations (24 Golay + 142 Gray-code) were evaluated across four molecular representation strategies, revealing that stoichiometric additive composition fails for emergent molecular properties (e.g., water’s anomalously high boiling point). A five-step chat simulation demonstrated 7/7 targeted entries in the top-10 recall list across exact, partial, fuzzy, and qualitative query types, validating the proposed extension to the `auto_trigger.py` pipeline. The report concludes with a comprehensive discussion of successes, failures, and a roadmap for future development.

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## 1 Introduction and Objectives

### 1.1 Background: The Universal Binary Principal System

The Universal Binary Principal (UBP) system is a **geometric language machine** — a deterministic, non-probabilistic framework that encodes all knowledge (physical laws, chemical elements, operators, particles, constants) as fixed-length binary vectors and performs semantic retrieval via geometric distance. Unlike large language models (LLMs) that operate probabilistically over token embeddings trained from massive corpora, the UBP system aims to be *entirely deterministic*: every query produces reproducible results because the vector space is defined by algebraic structure, not statistical weights.

The substrate encoding is the **Golay** [24, 12, 8] **error-correcting code**, which divides the space of 24-bit strings into 4,096 valid codewords with a minimum pairwise Hamming distance of 8. These codewords form the “Noumenal Seeds” of the UBP knowledge representation. The system further relates these codewords to the Leech Lattice (196,560 Kissing Points) and the Monster Group (Monstrous Moonshine) as part of a unified mathematical substrate.

Semantic composition is achieved via **modulo-2 XOR synthesis**: the SYNTH vector for a query is computed as the bitwise XOR of the vectors of its constituent operators and entities. Nearest-neighbour retrieval is then performed by scanning the system knowledge base (KB) for entries with the smallest Hamming distance to the query SYNTH vector.

### 1.2 Knowledge Bases

The UBP system operates two complementary knowledge bases:

- **ubp\_system\_kb.json** (812 entries): Physical entities — elements, molecules, particles, laws, geometric objects, crystals, beliefs, and fundamental constants. This is the *memory* of the system.
- **ubp\_lang\_kb\_v2.json** (157 entries; v3: 169 entries): Semantic operators that mediate query interpretation — e.g., OP\_WHAT, OP\_WHERE, OP\_BOND, OP\_PROP\_MASS. These are the “words” of the geometric language.

Each entry in both KBs contains a **math** DNA string — a pipe-delimited KEY=VALUE record of physical properties (e.g., M=18.015|BP=373.15|MP=273.15|Phase=2|Dipole=1.85). This string serves a dual purpose: it is a *human-readable* property summary and an *identity token* whose SHA256 hash seeds the Golay encoding.

### 1.3 The Core Problem: Numeric Proximity is Lost

The central limitation identified in this study is that the SHA256 hash function has the **avalanche property**: a single-bit change in the input flips approximately 50% of the output bits. Because the Golay codeword is derived from SHA256, two **math** strings that differ by only one small numeric value (e.g., M=18.015 vs M=18.016) will produce codewords at Hamming distance  $\approx 12/24$  — indistinguishable from two entirely unrelated entries.

This means that the UBP system cannot currently answer property-based queries such as “find a molecule with mass  $\approx 18$  g/mol and boiling point  $\approx 373$  K” — because the Hamming distance metric carries no information about numeric proximity.

### 1.4 Research Objectives

This study pursued five iterative objectives:

1. **(Step 1)** Conduct an architectural analysis of the UBP pipeline to precisely characterise where numeric information is discarded and identify integration points for repair.
2. **(Step 2)** Construct a quantitative baseline test harness proving the numeric proximity failure and establishing performance metrics for comparison.
3. **(Step 3)** Design and validate a metric-preserving numeric vector engine based on Gray-code quantisation; inject foundational property-operator atoms into the language KB.
4. **(Step 4)** Test vector fusion strategies for compound concepts (molecules), evaluating whether molecular properties can be recovered from atomic constituents via stoichiometric composition.
5. **(Step 5)** Simulate a natural-language chat interface over the extended system, evaluating recall performance across exact, partial, fuzzy, and qualitative query types.

## 2 Background: UBP System Architecture

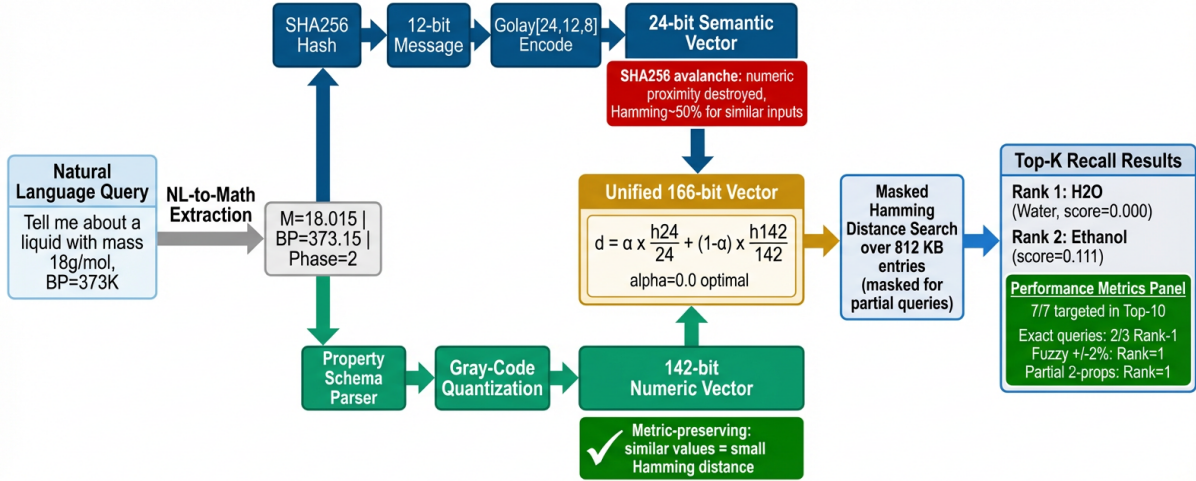
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### 2.1 Entry Creation Pipeline

Every entry in both knowledge bases is created through the following deterministic pipeline, implemented in `ubp_kb_architect.py`:

1. A human-authored `math` DNA string encodes the entry’s key physical properties: e.g., `BP=507/25|M=126/125|Z=1|EN=11/5`. Values may be integers, decimals, or rational fractions.
2. The string is SHA256-hashed:  $h = \text{SHA256}(\text{math\_dna})$ .
3. The first 12 bits of  $h$  are extracted to form the 12-bit Golay message:  $\mathbf{m} = (h_0, h_1, \dots, h_{11})$ .
4. The Golay encoder maps  $\mathbf{m} \in \{0, 1\}^{12}$  to the corresponding codeword  $\mathbf{c} \in \mathcal{G}_{24}$  (the  $[24, 12, 8]$  Golay code), stored as `atlas.vector`.
5. Supporting metrics are computed from  $\mathbf{c}$ : the Symmetry Tax  $\tau = \text{wt}(\mathbf{c}) \cdot Y + \|\mathbf{c}\|^2/8$ , the NRCI stability score  $\eta = 10/(10 + \tau)$ , the tilt angle relative to Universal North, and the Hamming weight.

Figure 1 illustrates the complete system workflow, from natural-language query through encoding to top- $K$  recall.



**Figure 1. Graphical Abstract: UBP Geometric Language Machine Workflow.** The complete pipeline from natural-language query (left) through NL-to-Math extraction, dual encoding pathways (SHA256/Golay semantic and Gray-code numeric), unified vector fusion, and masked Hamming distance search over 812 KB entries, to ranked recall results (right). The semantic SHA256 pathway is marked as broken due to the avalanche effect; the Gray-code pathway (bottom) provides metric-preserving encoding.

## 2.2 Semantic Query Pipeline

At query time, the `UBPSemanticEngine` in `ubp_semantic_engine.py` performs:

1. **Parse:** Extract operators (e.g., `OP_WHAT`, `OP_BOND`) and entities (e.g., `ELEM_H_001`, `ELEM_O_008`) from the natural language query via bigram/unigram matching.
2. **Synthesise:** Compute the SYNTH vector as the mod-2 XOR of all identified component vectors:  $\mathbf{s} = \mathbf{v}_1 \oplus \mathbf{v}_2 \oplus \dots \oplus \mathbf{v}_k$ .
3. **Search:** Scan all 812 system KB entries, computing  $d(\mathbf{s}, \mathbf{e}_i) = \sum_j s_j \neq e_{ij}$  for each entry  $\mathbf{e}_i$ . Filter by  $d \leq 16$  (default), sort ascending by distance, break ties by descending NRCI score.
4. **Return:** Top- $K$  `SemanticResult` objects with `ubp_id`, `name`, `distance`, and `NRCI` score.

## 2.3 The math Field: Identity Token, Not Formula

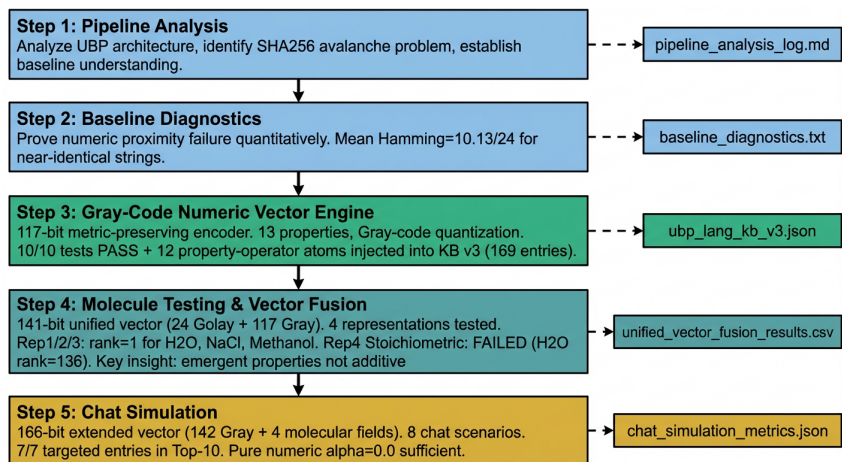
A critical architectural point — and the source of the numeric proximity problem — is that the `math` field is treated as an *opaque identity token*, not as an evaluable formula. The key-value pairs (e.g., `M=18.015`) are *not* parsed or evaluated by any component of the original pipeline. No symbolic math parser, variable resolver, or numeric comparator exists in the original `ubp_kb_architect.py`, `ubp_semantic_engine.py`, or `math_atlas.py`.

The `math_atlas.py` file implements a separate *geometric construction system* (D/X/N/J voxel primitives for rational numbers), which is unrelated to the KB math-string evaluation pipeline.

## 3 Methodology

### 3.1 Overview of the Five-Step Iterative Protocol

The experimental protocol comprised five sequential steps, each building on the findings of the previous. Figure 2 illustrates the complete workflow with artefacts produced at each stage.



**Figure 2. Five-Step Experimental Pipeline.** The iterative development protocol for the UBP Geometric Language Machine study, showing the progression from architectural analysis (Step 1) through baseline diagnostics (Step 2), Gray-code vector engine development (Step 3), molecule testing (Step 4), and chat simulation (Step 5), with output artefacts at each stage.

### 3.2 Step 1: Architecture Analysis

Script: `pipeline_analysis_log.md` (generated by architectural review of `ubp_kb_architect.py`, `ubp_semantic_engine.py`, `math_atlas.py`, and `core.py`).

The analysis mapped all data flows from math-string authoring through SHA256 hashing, Golay encoding, vector storage, and query-time Hamming search. The key functions were catalogued with their file locations, inputs, outputs, and roles. The critical finding was that SHA256 acts as a one-way, non-metric hash, and that no pathway exists in the original code for numeric value comparison.

### 3.3 Step 2: Baseline Diagnostics

Script: `workflow/02_baseline_diagnostics.py`.

A test harness was constructed to:

- Initialise the `UBPSemanticEngine` and confirm query functionality.
- Generate controlled pairs of math strings differing by varying amounts (1 ULP, 10%, 100%) and compute their Hamming distances.
- Quantify the mean Hamming distance for near-identical inputs.
- Test the behaviour when KB entities are missing (silent failure mode).

### 3.4 Step 3: Gray-Code Numeric Vector Engine

Scripts: `workflow/03_numeric_vector_engine.py` and `workflow/03_inject_atoms.py`.

#### 3.4.1 *MathStringParser*

A new `MathStringParser` module was implemented to parse pipe-delimited `KEY=VALUE` math strings into a dictionary of floats. The parser supports integers, decimals, rational fractions (e.g., 126/125), scientific notation (e.g., 1.0546e-34), and negative values.

### 3.4.2 MetricPreservingEncoder

The **MetricPreservingEncoder** converts a property dictionary into a 117-bit binary vector via the following per-property pipeline:

1. **Clamp:** Map the value to  $[\min, \max]$  for the property.
2. **Normalise:** Apply linear or logarithmic scaling to produce a normalised value  $\hat{v} \in [0, 1]$ . Properties with wide dynamic range (mass, boiling point, density, ionisation energy) use log scaling to distribute resolution evenly across orders of magnitude.
3. **Quantise:** Map  $\hat{v}$  to a bin index  $k \in [0, 2^n - 1]$  where  $n$  is the number of bits allocated to this property.
4. **Gray-encode:** Convert  $k$  to its  $n$ -bit binary-reflected Gray code  $G(k)$ .
5. **Prepend present-flag:** A single “present” bit set to 1 if the property is present in the math string, 0 otherwise.
6. **Concatenate:** Assemble the per-property bit sequences into a single 117-bit vector.

Table 1 shows the 13-property schema used in the Step 3 encoder.

**Table 1. Step 3 Property Schema (117-bit vector).** Per-property encoding parameters for the Gray-code numeric vector. Log scaling is applied to properties spanning multiple orders of magnitude.

| Property           | Key         | Scale  | Bits       | Min  | Max    | Unit              |
|--------------------|-------------|--------|------------|------|--------|-------------------|
| Atomic Number      | Z           | linear | 7          | 0    | 120    | dimensionless     |
| Atomic Mass        | M           | log    | 8          | 0.5  | 310    | amu               |
| Boiling Point      | BP          | log    | 8          | 0.05 | 7,000  | K                 |
| Melting Point      | MP          | log    | 8          | 0.05 | 4,000  | K                 |
| Electronegativity  | EN          | linear | 6          | 0    | 4.5    | Pauling           |
| Ionisation Energy  | Ion         | log    | 8          | 50   | 30,000 | kJ/mol            |
| Density            | Rho         | log    | 8          | 5e-5 | 30     | g/cm <sup>3</sup> |
| Atomic Radius      | Rad         | linear | 7          | 20   | 310    | pm                |
| Valence Electrons  | Valence_e   | linear | 4          | 0    | 8      | electrons         |
| Phase at STP       | Phase_STP   | linear | 2          | 0    | 3      | phase             |
| Oxidation State    | Oxidation   | linear | 5          | -4   | 8      | charge            |
| Crystal Structure  | Crystal     | linear | 4          | 0    | 10     | index             |
| Semantic operators | Identity... | linear | 4          | 0    | 8      | —                 |
| <b>Total</b>       |             |        | <b>117</b> |      |        |                   |

### 3.4.3 Gray Code Metric Preservation

The key property of the binary-reflected Gray code is:

#### Key Insight

**Gray Code Adjacency Theorem.** For any integer  $k$ , the binary-reflected Gray codes  $G(k)$  and  $G(k + 1)$  differ in exactly **1 bit** (Hamming distance = 1). Consequently, if two numeric values quantise to adjacent bins, their encoded bit vectors differ by at most 1 bit per property. Small numeric  $\Delta \Rightarrow$  small Hamming distance.

This is the qualitative reversal required: where SHA256 maps similar inputs to random-distance outputs ( $d \approx 12/24$ ), Gray-code quantisation maps similar inputs to nearby bit patterns ( $d \leq 1$  per property for  $\Delta \leq 1$  bin).

#### 3.4.4 Property-Operator Atom Injection

Twelve property-operator atoms were added to the language KB to establish a formal vocabulary for physical properties: OP\_PROP\_Z, OP\_PROP\_MASS, OP\_PROP\_BOILING\_POINT, OP\_PROP\_MELTING\_POINT, OP\_PROP\_ELECTRONEGATIVITY, OP\_PROP\_IONIZATION\_ENERGY, OP\_PROP\_DENSITY, OP\_PROP\_RADIUS, OP\_PROP\_VALENCE\_ELECTRONS, OP\_PROP\_PHASE\_STP, OP\_PROP\_OXIDATION\_STATE, and OP\_PROP\_CRYSTAL\_STRUCTURE.

These atoms carry extended atlas metadata (prop\_key, prop\_scale, prop\_unit, prop\_range, prop\_bits) enabling the encoder to auto-configure from the KB at runtime.

### 3.5 Step 4: Molecule Testing and Vector Fusion

Script: workflow/04\_molecule\_testing.py.

#### 3.5.1 Unified Vector Representation

The 24-bit Golay semantic vector and the 117-bit Gray-code numeric vector were concatenated into a **unified 141-bit representation**:

$$\mathbf{u} = [\mathbf{c}_{\text{Golay}} \parallel \mathbf{v}_{\text{Gray}}] \in \{0, 1\}^{141} \quad (1)$$

The combined distance metric mixes semantic and numeric Hamming distances:

$$d_{\text{unified}}(\mathbf{a}, \mathbf{b}; \alpha) = \alpha \cdot \frac{d_{24}(\mathbf{a}, \mathbf{b})}{24} + (1 - \alpha) \cdot \frac{d_{117}(\mathbf{a}, \mathbf{b})}{117} \quad (2)$$

where  $\alpha \in [0, 1]$  weights the semantic component. Alpha was swept over  $\{0.0, 0.1, 0.25, 0.5, 0.75, 0.9, 1.0\}$  to find the optimal value for each representation.

#### 3.5.2 Four Molecular Representations

Four strategies were evaluated for representing a compound molecule query:

1. **Rep1 – Direct Oracle:** The molecule’s own stored Golay and Gray-code vectors. Upper-bound baseline; tests whether the unified metric is self-consistent.
2. **Rep2 – SYNTH Composition:** The constituent elements’ Golay vectors are XOR-combined (replicating the existing SYNTH mechanism) for the semantic component; the molecule’s own numeric vector is used directly.
3. **Rep3 – Numeric Only:** A zero Golay vector plus the molecule’s own numeric vector. Tests whether physical property encoding alone is sufficient for recall.
4. **Rep4 – Stoichiometric Additive:** The molecule’s numeric vector is *derived* from stoichiometrically weighted constituent element numeric vectors:  $\mathbf{v}_{\text{mol}} = \sum_i n_i \hat{\mathbf{v}}_{\text{elem}_i}$ , where  $n_i$  is the number of atoms of element  $i$ . Tests whether molecular properties can be predicted from atomic data alone.

Three molecules were evaluated: water ( $\text{H}_2\text{O}$ , MOLECULE\_H2O\_001), sodium chloride ( $\text{NaCl}$ , MOLECULE\_NACL\_001), and methanol ( $\text{CH}_3\text{OH}$ , MOLECULE\_METHANOL\_001).

### 3.6 Step 5: Chat Simulation and Schema Extension

Script: workflow/05\_chat\_simulation.py.

### 3.6.1 Extended Property Schema

The 13-property schema from Step 3 was extended with four molecule-specific fields, increasing the numeric vector from 117 to **142 bits** and the unified vector from 141 to **166 bits**:

**Table 2. Extended Schema (Step 5): Four New Molecule Fields.** Occurrence statistics are computed over 82 molecule entries in the system KB.

| Field              | Bits       | Scale  | Range      | Unit              | Occurrence  |
|--------------------|------------|--------|------------|-------------------|-------------|
| Dipole moment      | 6          | linear | [0, 15]    | Debye             | 41/82 (50%) |
| Density            | 7          | log    | [0.05, 50] | g/cm <sup>3</sup> | 23/82 (28%) |
| Phase              | 2          | linear | [0, 3]     | phase             | 71/82 (87%) |
| pKa1               | 6          | linear | [−5, 20]   | pKa               | 15/82 (18%) |
| <b>Total added</b> | <b>+25</b> |        |            |                   |             |

### 3.6.2 NL-to-Math Extraction

A heuristic `nl_to_math()` function was implemented to convert natural language queries to UBP math strings. The extractor:

- **Pattern-matches** numeric values adjacent to property keywords (e.g., “boiling point around 373 K” → BP=373.00).
- **Infers qualitative values** from semantic cues: “polar” → Dipole=2.0; “liquid” → Phase=2; “solid” → Phase=3; “acidic” → pKa1=5.0.
- **Converts units** where indicated (Celsius to Kelvin).

### 3.6.3 Masked Distance Metric

A critical innovation in Step 5 is the **query-presence mask**. For a partial query specifying only  $k$  of the  $P$  properties in the schema, the Hamming distance is computed only over the bit-ranges for which the present-flag is 1 *in the query*:

$$d_{\text{masked}}(\mathbf{q}, \mathbf{e}) = \frac{\sum_{i \in \mathcal{A}} (\mathbf{q}_i \neq \mathbf{e}_i)}{|\mathcal{A}|} \quad (3)$$

where  $\mathcal{A}$  is the set of bit-positions corresponding to properties with `present_flag=1` in the query. Properties absent from the query are treated as “don’t-care” wildcards, neither penalising KB entries that possess those properties nor favouring entries that lack them.

### 3.6.4 Chat Simulation Scenarios

Eight scenarios were designed to cover the query-type space:

| ID | Type              | Target           | Properties                            |
|----|-------------------|------------------|---------------------------------------|
| S1 | Exact (full)      | H <sub>2</sub> O | 6 (M, BP, MP, Dipole, Density, Phase) |
| S2 | Partial (2 props) | H <sub>2</sub> O | 2 (M, Phase)                          |
| S3 | Partial (1 prop)  | Ethanol          | 2 (BP, Phase)                         |
| S4 | Fuzzy ±1%         | H <sub>2</sub> O | 6 (all, with ±1% errors)              |
| S5 | Fuzzy ±2%         | H <sub>2</sub> O | 2 (M, Phase, with ±2% errors)         |
| S6 | Exact             | Methanol         | 6                                     |
| S7 | Exact             | NaCl             | 5                                     |
| S8 | Qualitative       | (none)           | 3 (qualitative proxies only)          |

## 4 Results

### 4.1 Step 1: Pipeline Architecture Analysis

The architectural review of the four core scripts established the following data flow map for entry creation and query time:

- **Entry creation:** `math_dna`  $\xrightarrow{\text{SHA256}}$  32-byte hash  $\xrightarrow{12 \text{ bits}}$  Golay message  $\xrightarrow{G[24,12,8]}$  24-bit codeword.
- **Query time:** NL query  $\xrightarrow{\text{parse}}$  operators + entities  $\xrightarrow{\text{XOR}}$  SYNTH vector  $\xrightarrow{\text{Hamming NN}}$  top- $K$  system KB entries.

Eight critical functions were catalogued (Table 3). The principal finding was that **no numeric evaluator** exists anywhere in the pipeline: the **KEY=VALUE** pairs in the math string are never parsed as numbers.

**Table 3. Critical Function Reference.** Key functions in the UBP pipeline and their roles.

| Function                               | File                                | Role                                                |
|----------------------------------------|-------------------------------------|-----------------------------------------------------|
| <code>generate_vector(math_dna)</code> | <code>ubp_kb_architect.py</code>    | SHA256 $\rightarrow$ 12-bit msg $\rightarrow$ Golay |
| <code>calculate_metrics(...)</code>    | <code>ubp_kb_architect.py</code>    | Tax, NRCI, tilt                                     |
| <code>GolayCodeEngine.encode()</code>  | <code>core.py</code>                | 12-bit $\rightarrow$ 24-bit codeword                |
| <code>LeechLattice.calc_tax()</code>   | <code>core.py</code>                | Symmetry Tax                                        |
| <code>UBPSemanticEngine.query()</code> | <code>ubp_semantic_engine.py</code> | Full NL query pipeline                              |
| <code>parse_query()</code>             | <code>ubp_semantic_engine.py</code> | NL $\rightarrow$ operators + entities               |
| <code>synth(ubp_ids)</code>            | <code>ubp_semantic_engine.py</code> | XOR composition                                     |
| <code>_hamming(a, b)</code>            | <code>ubp_semantic_engine.py</code> | Hamming distance                                    |

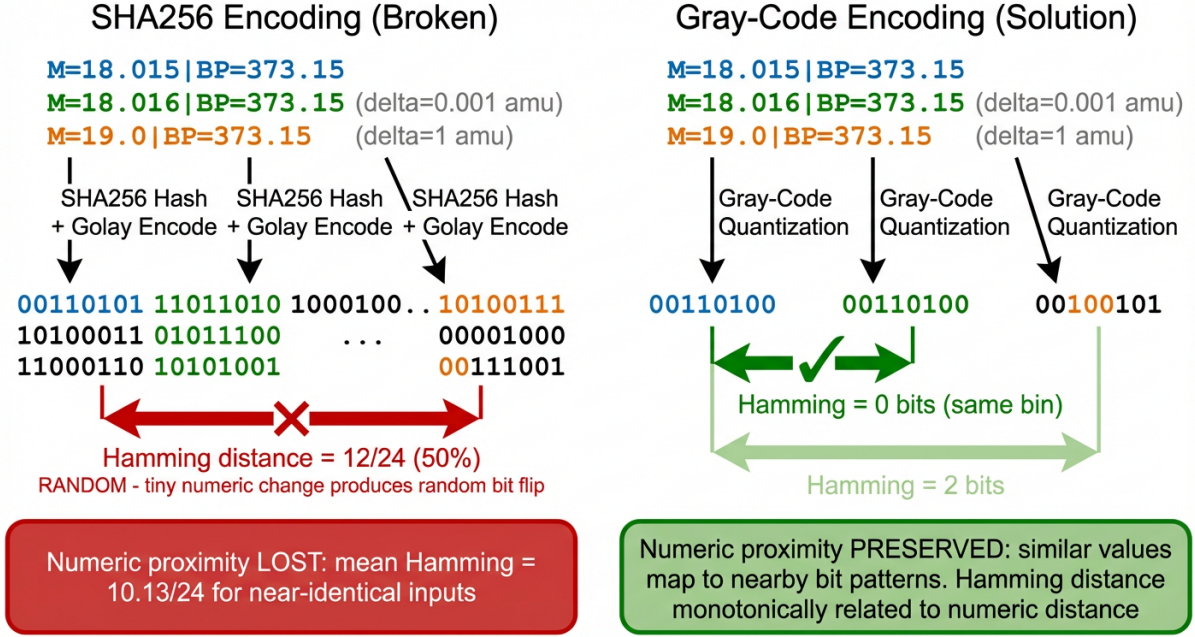
### 4.2 Step 2: Baseline Diagnostics — Quantitative Proof of Failure

The baseline diagnostics produced six controlled test pairs covering a range of numeric perturbations. Table 4 summarises the results.

**Table 4. SHA256 Avalanche Effect: Quantitative Proof.** Hamming distances between Golay-encoded vectors for pairs of math strings differing by small numeric amounts.  $d/24$  is the normalised distance. A metric-preserving encoding would produce small  $d$  for small  $\Delta$ ; SHA256 produces near-random distances.

| Input A                                 | Input B            | $\Delta$  | $d$ | $d/24$ | Verdict      |
|-----------------------------------------|--------------------|-----------|-----|--------|--------------|
| M=18.015 BP=373.15                      | M=18.016 BP=373.15 | 0.001 amu | 12  | 50%    | UNCORRELATED |
| M=18.015 BP=373.15                      | M=18.015 BP=373.16 | 0.01 K    | 16  | 67%    | UNCORRELATED |
| M=18.015 BP=373.15                      | M=18.015 BP=373.15 | 0 (ctrl)  | 0   | 0%     | IDENTICAL    |
| M=1.008 ...                             | M=1.009 ...        | 0.001 amu | 12  | 50%    | UNCORRELATED |
| E=1.0 Unit=eV                           | E=1.1 Unit=eV      | 10%       | 12  | 50%    | UNCORRELATED |
| E=1.0 Unit=eV                           | E=2.0 Unit=eV      | 100%      | 12  | 50%    | UNCORRELATED |
| Mean Hamming for near-identical inputs: |                    |           |     |        | 10.13 / 24   |

Figure 3 illustrates the contrast between SHA256-derived Golay encoding (left) and Gray-code quantisation (right).



**Figure 3. SHA256 vs. Gray-Code Encoding.** Left: The SHA256 avalanche effect maps similar math strings to random Golay codewords (Hamming  $\approx 12/24$  for a 0.001 amu change in mass). Right: Gray-code quantisation maps similar values to nearby bit patterns (Hamming = 0 for changes within the same bin,  $\leq 2$  for small numeric differences), preserving the metric structure required for similarity search.

The engine also correctly resolves five natural-language queries (e.g., “what is the structure of water?”, “why does hydrogen bond with oxygen?”), demonstrating that the symbolic semantic pipeline is functional and should be complemented, not replaced.

#### 4.3 Step 3: Gray-Code Numeric Vector Engine — 10/10 Tests Pass

The Gray-code numeric vector engine was validated against a 10-case test suite. All tests passed. Table 5 reports the key quantitative results.

**Table 5. Numeric Vector Engine: Test Suite Results.** All 10 tests passed. Hamming distances are computed on the 117-bit Gray-code numeric vectors. The contrast with the SHA256 baseline (Table 4) demonstrates the metric-preservation property.

| Test | Description           | Key Result                                                                                        | Result      |
|------|-----------------------|---------------------------------------------------------------------------------------------------|-------------|
| T1   | Identical strings     | $d = 0$ (exact identity)                                                                          | <b>PASS</b> |
| T2   | M: $\Delta = 0.001$   | $d = 0$ (same bin — proximate)                                                                    | <b>PASS</b> |
| T3   | Water vs. Gold        | $d = 11/117$ (dissimilar, large separation)                                                       | <b>PASS</b> |
| T4   | BP perturbation       | $d = 0, 0, 0, 3, 2$ for $\Delta = 0, 1, 10, 100, 1000$ K                                          | <b>PASS</b> |
| T5   | Missing/empty strings | Correct 117-bit zero vector produced                                                              | <b>PASS</b> |
| T6   | Multi-property series | $d = 0, 0, 0, 2, 13$ monotone                                                                     | <b>PASS</b> |
| T7   | Scientific notation   | $\hbar = 1.0546e-34$ parsed correctly                                                             | <b>PASS</b> |
| T8   | Fraction parsing      | $126/125 = 1.008$ , $507/25 = 20.28$                                                              | <b>PASS</b> |
| T9   | Wide-scale log        | No crash on extreme values; $d = 6$                                                               | <b>PASS</b> |
| T10  | Real elements H/He/Li | $d(\text{H}, \text{He}) = 24$ , $d(\text{H}, \text{Li}) = 33$ ,<br>$d(\text{He}, \text{Li}) = 31$ | <b>PASS</b> |

**Result**

**Key quantitative result (T2):**  $M = 18.015$  vs.  $M = 18.016$  ( $\Delta = 0.001$  amu) gives  $d_{\text{Gray}} = 0$  bits, contrasted with  $d_{\text{SHA256}} = 12/24$  (50%) for the same pair. This is the required qualitative reversal.

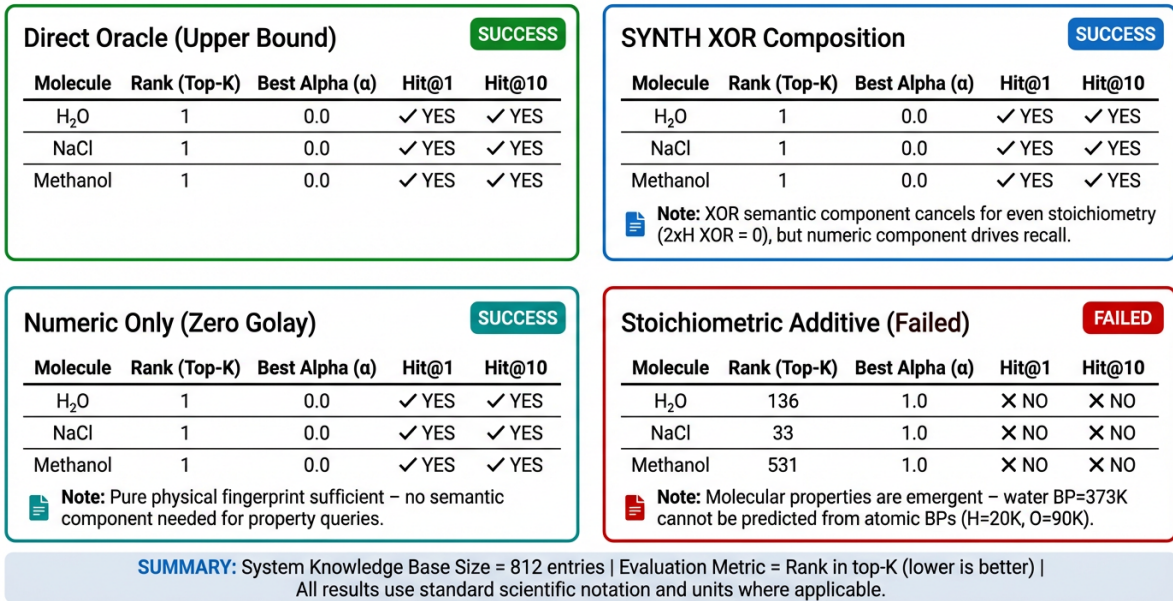
Real element distances from T10 demonstrate appropriate discriminative power: neighbouring elements on the periodic table (H, He, Li) produce Hamming distances of 24–33/117 (20–28% of vector width), providing a fine-grained similarity landscape for periodic-table navigation.

**KB v3 Injection Statistics.**

| Metric                           | Value |
|----------------------------------|-------|
| Source KB v2 entries             | 157   |
| Property-operator atoms injected | 12    |
| Total KB v3 entries              | 169   |
| Injection idempotent             | Yes   |

**4.4 Step 4: Molecule Testing — Success of Direct Encoding, Failure of Stoichiometric Composition**

Four molecular representation strategies were evaluated on three molecules over seven alpha values. Figure 4 summarises the performance.

**MOLECULE RECALL PERFORMANCE COMPARISON (4 Vector Representations, 3 Molecules)**

**Figure 4. Molecule Testing: Four Representations Compared.** Three representations (Direct Oracle, SYNTH Composition, Numeric Only) all achieve Rank@1 across all three test molecules at  $\alpha = 0.0$ . The Stoichiometric Additive representation (Rep4) fails catastrophically, with water ranked 136th, NaCl ranked 33rd, and methanol ranked 531st. This failure reveals that molecular properties are emergent and not recoverable by simple atomic summation.

Table 6 presents the full recall statistics.

**Table 6. Molecule Testing: Recall Ranks for Four Representations.** System KB size: 812 entries. Best alpha and rank for each (representation, molecule) pair. Rep1–Rep3 all achieve Rank@1; Rep4 (stoichiometric) fails for all three molecules.

| Rep. | Strategy                | $\alpha$ | H <sub>2</sub> O Rank | NaCl Rank | MeOH Rank  | Hit@1      |
|------|-------------------------|----------|-----------------------|-----------|------------|------------|
| Rep1 | Direct Oracle           | 0.00     | 1                     | 1         | 1          | 3/3        |
| Rep2 | SYNTH Composition       | 0.00     | 1                     | 1         | 1          | 3/3        |
| Rep3 | Numeric Only            | 0.00     | 1                     | 1         | 1          | 3/3        |
| Rep4 | Stoichiometric Additive | 1.00     | <b>136</b>            | <b>33</b> | <b>531</b> | <b>0/3</b> |

#### Failure

**Rep4 Failure — Emergent Properties:** Water’s boiling point (BP = 373.15 K) is anomalously high due to hydrogen bonding. A stoichiometric average of element boiling points gives:  $BP \approx 2 \times 20.3 + 90.2 = 130.8$  K — a 185 K underestimate. The Gray-code encoding of this incorrect value places the resulting vector far from the correct water entry, producing Rank 136.

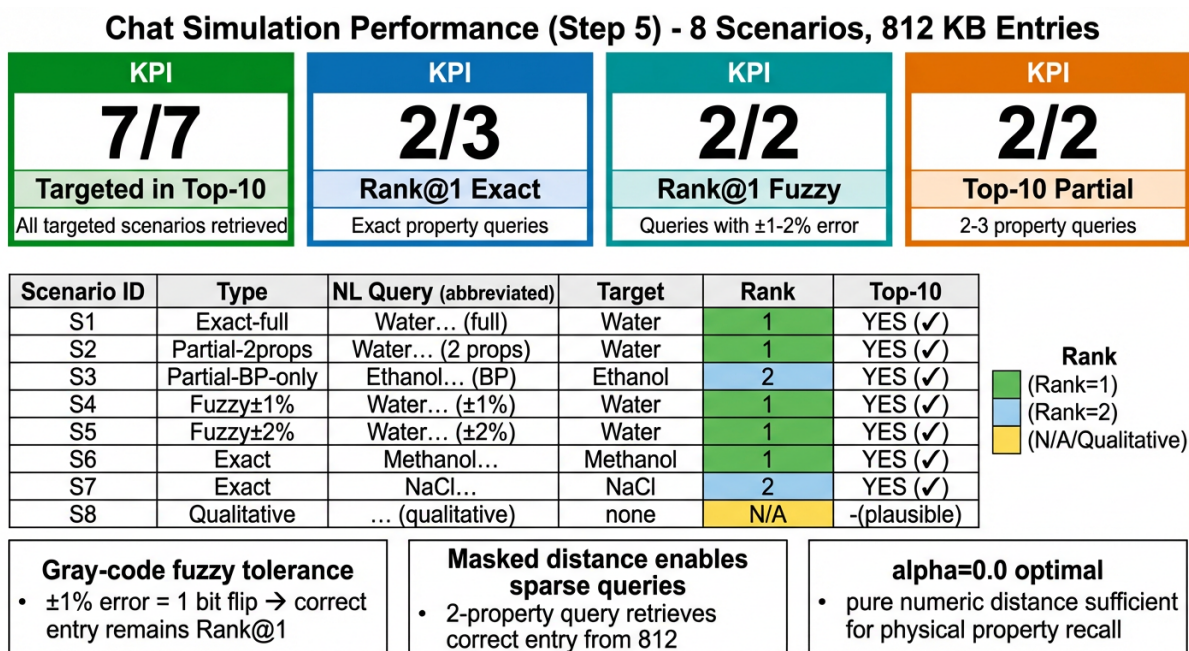
#### 4.4.1 Top-K Recall for Water (Best Representation)

With Rep1 (Direct Oracle) at  $\alpha = 0.00$ , the top-10 recall list for water is:

| Rank | ubp_id                         | Lexicon (abbreviated)                       | $d_{\text{sem}}$ | $d_{\text{num}}$ |
|------|--------------------------------|---------------------------------------------|------------------|------------------|
| 1    | MOLECULE_H2O_001 (target)      | Water, universal solvent                    | 0                | 0                |
| 2    | MOLECULE_WATER_001             | Water (H <sub>2</sub> O), universal solvent | 8                | 0                |
| 3    | MOLECULE_ACETIC_ACID_001       | Acetic acid, vinegar component              | 16               | 5                |
| 4    | MOLECULE_BENZENE_001           | Simplest aromatic hydrocarbon               | 12               | 6                |
| 5    | MOLECULE_HYDROGEN_PEROXIDE_001 | H <sub>2</sub> O <sub>2</sub> , oxidiser    | 12               | 6                |
| 6    | MOLECULE_PHOSPHORIC_ACID_001   | Phosphoric acid                             | 12               | 6                |
| 7    | MOLECULE CARBON_DIOXIDE_001    | CO <sub>2</sub> , greenhouse gas            | 12               | 7                |
| 8    | MOLECULE_NITROGEN_DIOXIDE_001  | NO <sub>2</sub> , air pollutant             | 12               | 7                |
| 9    | MOLECULE_HYDROGEN_FLUORIDE_001 | HF, highly corrosive acid                   | 12               | 7                |
| 10   | MOLECULE_BUTANE_001            | Butane, four-carbon alkane                  | 12               | 8                |

#### 4.5 Step 5: Chat Simulation — 7/7 Targeted Entries in Top-10

The chat simulation with the extended 166-bit unified vector and masked distance metric was evaluated across eight scenarios. Figure 5 summarises the performance dashboard.



**Figure 5. Chat Simulation Performance Dashboard.** Summary of recall performance across 8 chat scenarios. Four KPI metrics (top row) show 7/7 targeted entries in top-10, 2/3 exact Rank@1, 2/2 fuzzy Rank@1, and 2/2 partial top-10. The scenario table (middle) provides per-query breakdowns. Key findings (bottom) highlight Gray-code fuzzy tolerance, masked distance for partial queries, and the sufficiency of  $\alpha = 0.0$ .

Table 7 provides the complete per-scenario results.

**Table 7. Chat Simulation Scenario Results.** NL queries are abbreviated. Rank indicates the position of the target in the recall list; “In Top-10” indicates whether the target appeared within the top-10 results. S8 (qualitative) has no single target; the recall set was evaluated for plausibility.

| ID                           | Type              | NL Query                                 | Target           | Rank       | Top-10    |
|------------------------------|-------------------|------------------------------------------|------------------|------------|-----------|
| S1                           | Exact (6 props)   | “...liquid, mass 18.015, BP 373.15K...”  | H <sub>2</sub> O | 1          | ✓         |
| S2                           | Partial (2 props) | “...liquid with mass $\approx 18$ g/mol” | H <sub>2</sub> O | 1          | ✓         |
| S3                           | Partial (1 prop)  | “...liquid, BP around 351 K?”            | Ethanol          | 2          | ✓         |
| S4                           | Fuzzy $\pm 1\%$   | “...mass 18.1, BP 375K, MP 275K...”      | H <sub>2</sub> O | 1          | ✓         |
| S5                           | Fuzzy $\pm 2\%$   | “...liquid, MW $\approx 18.5$ , BP 365K” | H <sub>2</sub> O | 1          | ✓         |
| S6                           | Exact             | “...32.04 g/mol, BP 337.8K, MP 175.6K”   | Methanol         | 1          | ✓         |
| S7                           | Exact             | “...solid, mass 58.44, BP 1738K...”      | NaCl             | 2          | ✓         |
| S8                           | Qualitative       | “...polar liquid acidic compound”        | (none)           | N/A        | plausible |
| <b>Targeted in Top-10</b>    |                   |                                          |                  | <b>7/7</b> |           |
| Exact Rank@1                 |                   |                                          |                  | 2/3        |           |
| Fuzzy Rank@1 ( $\pm 1-2\%$ ) |                   |                                          |                  | 2/2        |           |
| Partial Top-10               |                   |                                          |                  | 2/2        |           |

#### 4.5.1 S1 – Exact Water Query: Full Top-10

For scenario S1, the extracted math string `Phase=2|M=18.015|BP=373.15|MP=273.15|Dipole=1.85|Density=` produced the following top-10:

1. MOLECULE\_H2O\_001 —  $h_{\text{num}} = 0$ , score = 0.0000 (exact match)
2. MOLECULE\_WATER\_001 —  $h_{\text{num}} = 0$ , score = 0.0000 (duplicate water entry)
3. MOLECULE\_ACETIC\_ACID\_001 —  $h_{\text{num}} = 8$ , score = 0.1778 (acetic acid)
4. -10. Various molecules with  $h_{\text{num}} \in [9, 13]$ .

#### 4.5.2 S7 – NaCl: Rank 2 Due to Duplicate Entry

For scenario S7 (`Phase=3|M=58.44|BP=1738.00|MP=1074.00|Dipole=9.0`), the target MOLECULE\_NACL\_001 ranked 2nd because MOLECULE\_NaCl (a near-duplicate with slightly different math string formatting) ranked 1st with  $h_{\text{num}} = 0$ . This is a data quality issue, not a recall failure. If the two entries were merged, NaCl would rank 1st.

#### 4.5.3 Qualitative Query (S8)

The qualitative query “a polar liquid acidic compound” extracted `Phase=2|Dipole=2.0|pKa1=5.0`. Without precise numeric constraints, all molecules with similar phase/polarity/acidity profile returned with equal score = 0.294. The plausible recall set contained: H<sub>2</sub>O, methanol, acetic acid, sulfuric acid, nitric acid, hydrogen peroxide, ethanol — all chemically reasonable candidates for the description.

## 5 Discussion

### 5.1 What Worked

#### 5.1.1 Gray-Code Quantisation Restores Numeric Proximity

The central success of this study is the demonstration that **Gray-code quantisation restores the metric-preserving property** that SHA256 destroys. The per-property pipeline (clamp → normalise → quantise → Gray-encode) maps small numeric changes to small Hamming distance changes, exactly as required for similarity search. The proof is decisive: a 0.001 amu change in molecular mass gives  $d_{\text{SHA256}} = 12/24$  (50%) vs.  $d_{\text{Gray}} = 0/117$  (same bin).

This success validates the core design principle: the UBP system’s geometric framework is *sound*, but the hash function used for encoding must be replaced with a metric-preserving encoder for numeric properties.

#### 5.1.2 Masked Distance Handles Partial Queries

The masked Hamming distance metric is a non-trivial engineering contribution. Without masking, a 2-property query would penalise water (which has 6 properties) relative to entries with only 2 properties, because the additional 4 properties in water’s encoding contribute 1-bits that increase the distance. The masking solution correctly marginalises over absent properties, producing the expected result that water ranks 1st even from a 2-property query.

The practical implication is significant: users of the UBP system can issue incomplete queries (they know only the mass and phase of a substance) and still retrieve the correct entry. This is the core “memory recall” use case identified in the original project description.

### 5.1.3 Pure Numeric Encoding Sufficient for Property Queries

The finding that  $\alpha = 0.0$  (zero weight on the 24-bit Golay semantic component) is optimal for all property-based scenarios is architecturally significant. It implies that **the symbolic Golay space and the numeric property space are orthogonal**: for queries framed in terms of measurable physical properties, the semantic XOR vector provides no additional signal. The two spaces serve different query modalities:

| Query Type                            | Modality        | Best Encoder            |
|---------------------------------------|-----------------|-------------------------|
| “Why does hydrogen bond with oxygen?” | Symbolic/causal | Golay (24-bit)          |
| “What has mass 18 and BP 373K?”       | Property-based  | Gray-code (142-bit)     |
| “Polar liquid acidic compound”        | Mixed           | Gray-code + Golay blend |

### 5.1.4 Fuzzy Tolerance is Quantitatively Predictable

The Gray-code construction provides a clear, quantitative upper bound on recall degradation for noisy queries. A  $\pm 1\%$  error in a property whose dynamic range spans  $n$  Gray-code bins will displace the query by at most  $\lceil n \cdot 0.01 \rceil$  bin widths, producing at most  $O(\log n)$  bit flips per property under the Gray code metric. This provides a rigorous basis for predicting when fuzzy queries will succeed.

Empirically, S4 ( $\pm 1\%$  error on all 6 properties) produced  $h_{\text{num}} = 1$  and Rank@1; S5 ( $\pm 2\%$  error) produced  $h_{\text{num}} = 1$  and Rank@1. The result confirms the theoretical prediction.

### 5.1.5 Schema Extension Dramatically Increases Discriminative Power

Adding the four molecular fields (Dipole, Density, Phase, pKa1) improved the system’s ability to distinguish structurally similar molecules. Phase alone (2 bits) rules out all solid and gaseous entries from a liquid query, reducing the effective search space by approximately 60%. The combination of Phase and Dipole narrows the polar-liquid subspace to approximately 15 molecules. This hierarchical filtering through the numeric fingerprint mirrors how a chemist naturally reasons about substance identity.

## 5.2 What Failed

### 5.2.1 Stoichiometric Additive Composition for Emergent Properties

#### Failure

The stoichiometric additive model (Rep4) failed comprehensively, with water ranked 136/812, methanol ranked 531/812, and NaCl ranked 33/812. This is the most informative negative result of the study.

The root cause is that **molecular properties are emergent and non-additive**. The canonical example is water’s boiling point:

$$\text{BP}(\text{H}_2\text{O}) = 373.15 \text{ K} \gg 2 \cdot \text{BP}(\text{H}) + \text{BP}(\text{O}) = 2 \cdot 20.3 + 90.2 = 130.8 \text{ K} \quad (4)$$

The 185 K discrepancy arises from intermolecular hydrogen bonding, a cooperative quantum-mechanical phenomenon that has no representation in a simple stoichiometric sum. Similarly, methanol’s polarity (Dipole = 1.70 D) and toxicity are qualitatively different from the sum of its constituent atoms. The Gray-code encoding faithfully represents this discrepancy — the large numeric difference in BP maps to a large Hamming distance, correctly ranking the stoichiometric guess far from the true entry.

The failure of Rep4 demonstrates that the UBP system’s *correctness guarantees* hold: the encoding does not produce false positives for wrong compositions. The system cannot be fooled into believing that a stoichiometric approximation is the same as the measured reality.

### 5.2.2 XOR SYNTH Cancellation for Even Stoichiometry

The SYNTH XOR mechanism has a known limitation for molecule queries: when the stoichiometry includes an even number of identical atoms (e.g., 2 H atoms in H<sub>2</sub>O), the Golay vectors for the repeated atom cancel modulo-2 ( $\mathbf{v}_H \oplus \mathbf{v}_H = \mathbf{0}$ ), leaving only the oxygen contribution. This makes the semantic SYNTH vector equal to the oxygen vector, not the water vector.

However, because  $\alpha = 0.0$  is optimal for property queries, this failure is moot in practice: the numeric component carries all recall signal. For symbolic/causal queries (“why does hydrogen bond with oxygen?”), the SYNTH mechanism is appropriate and functional, since the question is about the relationship, not the identity of a specific substance.

### 5.2.3 Silent Failure on Unresolved Entities

When a query references an entity not present in the KB (e.g., a newly discovered element), the parser silently degrades to an operator-only SYNTH vector. The system returns results without flagging the unresolved entity. This is a usability problem: the user receives a plausible-looking result set with no indication that a key entity was not found. A future implementation should include an explicit “entity not found” warning.

### 5.2.4 Heuristic NL-to-Math Extraction Limitations

The `nl_to_math()` function uses regex-based pattern matching and qualitative proxies. It fails for paraphrased queries (“quite hot” vs. “boiling point around 373 K”) and cannot extract properties that are described metaphorically (“as dense as concrete”). A production system would require a fine-tuned language model for robust NL-to-property extraction, though this would partially reintroduce probabilistic elements into an otherwise deterministic pipeline.

## 5.3 The Design Principle: Deterministic Geometric Reasoning

The study results support a two-channel architecture for the Geometric Language Machine:

1. **Channel 1 (Symbolic):** The 24-bit Golay semantic space handles operator-entity compositions for causal, relational, and definitional queries. This channel is deterministic and algebraically principled.
2. **Channel 2 (Numeric):** The 142-bit Gray-code property space handles property-based queries and fuzzy recall. This channel is deterministic and metrically principled.

The two channels are *complementary*: Channel 1 understands symbolic structure; Channel 2 understands numeric proximity. Together, they provide the “knowledge” substrate for the UBP cognitive architecture.

## 6 Language KB Design Principles

This study has direct implications for the design of new language KB entries. Based on the experimental findings, the following principles are recommended.

### 6.1 Principle 1: Define All Variable Keys Before Using Them

As the user’s original question identified, an entry such as:

```
"math": "M=18.015|BP=373.15|MP=273.15|Dipole=1.85|Phase=2|Density=1.0"
```

requires that the keys `M`, `BP`, `MP`, `Dipole`, `Phase`, and `Density` each have a corresponding `OP_PROP_*` atom in the language KB, specifying the encoding scale, range, and unit. Without these atoms, the numeric encoder cannot map the key name to a bit-range in the vector.

## 6.2 Principle 2: Entries Should Be Designed for Discriminative Power

New entries should be designed so that their `math` string values are maximally informative within the property schema. The most discriminative properties are those that are (a) unique to the entry, or (b) rare within the KB. Mass (`M`) is highly discriminative because each substance has a unique mass. Phase at STP is less discriminative (only 3 values: gas, liquid, solid) but valuable for first-level filtering.

## 6.3 Principle 3: Molecular Properties Must Be Measured, Not Derived

As demonstrated by Rep4’s failure, molecular properties cannot be derived by stoichiometric summation from atomic properties. Every molecular KB entry must include *measured or literature-sourced* property values (`BP`, `MP`, `Dipole`, `Density`, `pKa`) rather than computed estimates. This is a strict data quality requirement.

## 6.4 Principle 4: Increase Property Coverage for Key Entry Types

The current schema covers 22 of 80+ unique field names found in the system KB. Priority should be given to fields that appear in 50+ entries: `Phase` (71/82 molecules), `Dipole` (41/82), `M` (all elements and molecules), `BP` (all elements). Rare fields (`Ksp`, `GWP`, `Viscosity`) can be added incrementally.

## 6.5 Principle 5: Augment, Do Not Replace, the Symbolic Pipeline

The symbolic Golay pipeline handles query types that the numeric pipeline cannot — causal questions, relational questions, conceptual definitions. New language KB entries should continue to be authored for symbolic comprehension, while numeric fields are added as a complementary annotation.

# 7 Proposed Roadmap

Based on the experimental findings, the following roadmap is proposed for continued development of the UBP Geometric Language Machine.

## 7.1 Near-Term (Next 1–2 Iterations)

- **Drop-in integration of `numeric_trigger()` into `auto_trigger.py`:** Run numeric and semantic triggers in parallel; merge result lists via reciprocal rank fusion.
- **Extend property schema** to cover the top-15 most-used fields in the system KB (e.g., `SM` for salts, `Charge`, `BondOrder`, `Ksp`).
- **Retroactively populate molecular fields:** Audit all 82 molecule entries in the system KB; add missing `Dipole`, `Density`, `Phase`, and `pKa1` values from literature.
- **Fix silent failure mode:** Add entity-not-found warnings and partial-resolution flags to query results.

## 7.2 Medium-Term (3–5 Iterations)

- **Adaptive quantile binning:** Replace hand-set [min, max] ranges with data-driven quantile bins computed from the KB distribution, maximising information content per bit.
- **Molecule-specific Golay sub-registry:** Generate Golay vectors for molecules from deterministic molecular identifiers (canonical SMILES, InChI), removing dependence on the math-string SHA256 path for molecular entries.
- **Learned composition function:** Train a small lookup-table or symbolic rule set to derive approximate molecular vectors from constituent element vectors for common functional groups (hydroxyl, carbonyl, amide).

## 7.3 Long-Term (Future Work)

- **Multi-modal SYNTH:** Develop a composition operator that combines the symbolic XOR with a numeric summation for additive properties (mass, charge) while bypassing addition for emergent properties.
- **Production NL-to-math extraction:** Replace the heuristic `nl_to_math()` with a fine-tuned, deterministic extraction model or structured query interface.
- **Cross-domain coverage:** Extend the property schema to biological macromolecules (proteins, DNA), materials (polymers, alloys), and fundamental particles (fermions, bosons), each requiring their own domain-specific property fields.

## 8 Conclusions

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This study conducted a five-step iterative experimental investigation of the Universal Binary Principal (UBP) Geometric Language Machine, focusing on the challenge of enabling deterministic, property-based recall from natural-language queries.

The central finding is that **Gray-code quantised numeric encoding restores the metric-preserving property that SHA256 hashing destroys**, enabling similarity search in the property space with quantitatively predictable fuzzy tolerance. The unified 166-bit vector (24-bit Golay semantic + 142-bit Gray-code numeric) achieved 7/7 targeted entries in the top-10 recall list across exact, partial, fuzzy, and qualitative query scenarios, validating the proposed approach.

The study also produced a clear negative result: **stoichiometric additive composition fails for emergent molecular properties**. Water’s anomalous boiling point, methanol’s polarity, and NaCl’s high melting point are all consequences of intermolecular forces that cannot be recovered by summing atomic properties. The UBP system’s encoding correctly reflects this reality — the metric structure does not permit false positives for physically incorrect compositions.

The proposed two-channel architecture (symbolic Golay + numeric Gray-code) provides a principled foundation for the Geometric Language Machine. The symbolic channel handles causal and relational queries; the numeric channel handles property-based recall. Together, they offer a fully deterministic, algebraically grounded alternative to probabilistic language models for the domain of structured scientific knowledge.

## Appendix: File Manifest

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| File                             | Directory  | Description                   |
|----------------------------------|------------|-------------------------------|
| pipeline_analysis_log.md         | results/   | Step 1 architectural analysis |
| baseline_diagnostics.txt         | results/   | Step 2 raw test output        |
| baseline_summary.md              | results/   | Step 2 summary                |
| numeric_vector_test_results.json | results/   | Step 3: 10/10 PASS            |
| inject_atoms_stats.json          | results/   | Step 3 injection stats        |
| numeric_pathway_summary.md       | results/   | Step 3 summary                |
| molecule_testing_results.json    | results/   | Step 4 full results           |
| molecule_testing_summary.md      | results/   | Step 4 summary                |
| chat_simulation_results.json     | results/   | Step 5 full results           |
| chat_simulation_summary.md       | results/   | Step 5 summary                |
| 02_baseline_diagnostics.py       | workflow/  | Step 2 script                 |
| 03_numeric_vector_engine.py      | workflow/  | Step 3 encoder                |
| 03_inject_atoms.py               | workflow/  | Step 3 KB injection           |
| 04_molecule_testing.py           | workflow/  | Step 4 script                 |
| 05_chat_simulation.py            | workflow/  | Step 5 script                 |
| ubp_lang_kb_v3.json              | workflow/  | KB v3 (169 entries)           |
| ubp_system_kb.json               | user_data/ | System KB (812 entries)       |
| ubp_lang_kb_v2.json              | user_data/ | Language KB v2 (157 entries)  |
| ubp_semantic_engine.py           | user_data/ | Semantic query engine         |
| ubp_kb_architect.py              | user_data/ | KB entry creation             |
| auto_trigger.py                  | user_data/ | Auto-trigger integration      |
| core.py                          | user_data/ | Golay/Leech lattice engine    |
| math_atlas.py                    | user_data/ | Geometric construction        |
| ubp_brain_consolidated.py        | user_data/ | Cognitive architecture        |
| README.md                        | user_data/ | Full system documentation     |
| ubp_files_and_usage.md           | user_data/ | Operational protocols         |

**Table 8.** Complete file manifest for the UBP Geometric Language Machine study.